Please write your Enrollment Number) Enrollment No. 03201121021

END-TERMEXAMINATION

(CBCS)

B. Tech (CSE/ECE), Semester: II

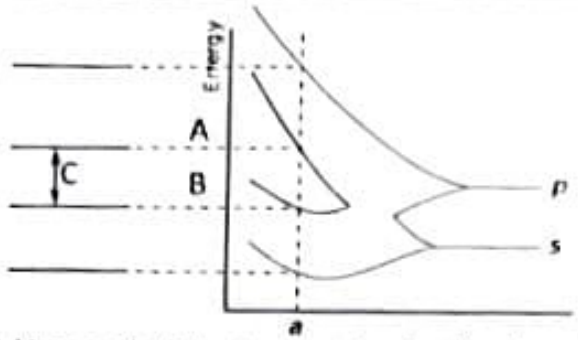
(AUGUST, 2022)

Subject Code: BAS-104	Subject: Applied Physics - II

Time :3 Hours Maximum Marks :60

Note:Q. 1 is compulsory. (Do any four questions out of five).
Attempt one question from each unit.

Q1		(5*4=20)
	(a) Charges are distributed in a volume non uniformly with a charge density given by $\rho(r,\theta,\phi)$. The medium has relative permittivity ϵ_r. Write the expression for Gauss's law for these charges.	
	(b) Calculate the skin depth for a frequency of 10^{18}Hz for silver if $\mu_o = 4\pi \times 10^{-7}$ Weber/A-m, $\sigma = 3 \times 10^7$ S/m.	
	(c) Draw four separate cubes and indicate the three-unit vectors as (a_1, a_2, a_3). Now, shade the following lattice planes in each one of them (1 0 1), ($\bar{1}$ 0 1),(0 1 2), (2 2 1).	
	(d) Prove that electromagnetic waves propagate with the speed of light in free space?	
	(e) What is the difference between a sensor, transducer, and an actuator? Give examples of each	
	UNIT-1	
Q2		(5,5)
	(a) Derive an expression for the propagation of electromagnetic waves in a dielectric medium. (b) What is current density? How is it different from current? Determine the displacement current for a parallel place capacitor, with radius 0.055m, and $\frac{\partial E}{\partial t} = 1.5 \times 10^{12}\ V/ms$.	
	OR	
Q3		(5,5)
	(a) Explain how does the Fermi level change for a p and n type semiconductor. Draw a suitable diagram. (b) What is displacement current? How the modified Maxwell's equation established consistency in the theory of propagation of EM waves?	
	UNIT-2	
Q4		(5,5)(10)
	Discuss E Vs k diagram for a one-dimensional crystal lattice. Explain the reason why certain values of k are forbidden?	
	OR	

Q5		(7,3)
	 (a)Answer the following questions for the above diagram: I. What does these curves it represent? II. What quantity is on the x-axis. III. What does 'a' represent in this plot. IV. What does Energy level A and B denote? V. Which quantity does C represent. VI. What is the reason for transition from right to left? (b) Write the expression for Bloch Theorem.	
	UNIT-3	
Q6	(a) Explain transfer and inverse transfer function of a sensor using different examples? Also Explain the terms like Span, full scale output, dead band, and saturation. **(5)** (b) Explain the term pre-aging. Why is it required for a sensor? How is it done? What is the accelerated life tests? Also list four environmental tests. **(5)**	(5,5)
	OR	
Q7	(a) Explain what should be the characteristics of an ideal sensor? **(5)** (b) Give the construction and working of a capacitative water level sensor. (5)	(5,5)
	UNIT-4	
Q8	Why electrons are preferred for microscopic technique such as SEM (Scanning electron Microscope). Explain the principle, construction and working of SEM with the help of a diagram? (10)	(10)
	OR	
Q9	(a) How is the third generation CT scanner different from first and second generation.**(5)** (b) Explain the following terms for a CT scan(i) Sinogram(ii) Radon Transform(iii)CT number. **(5)**	(5,5)